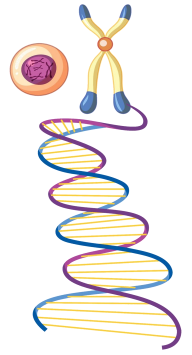


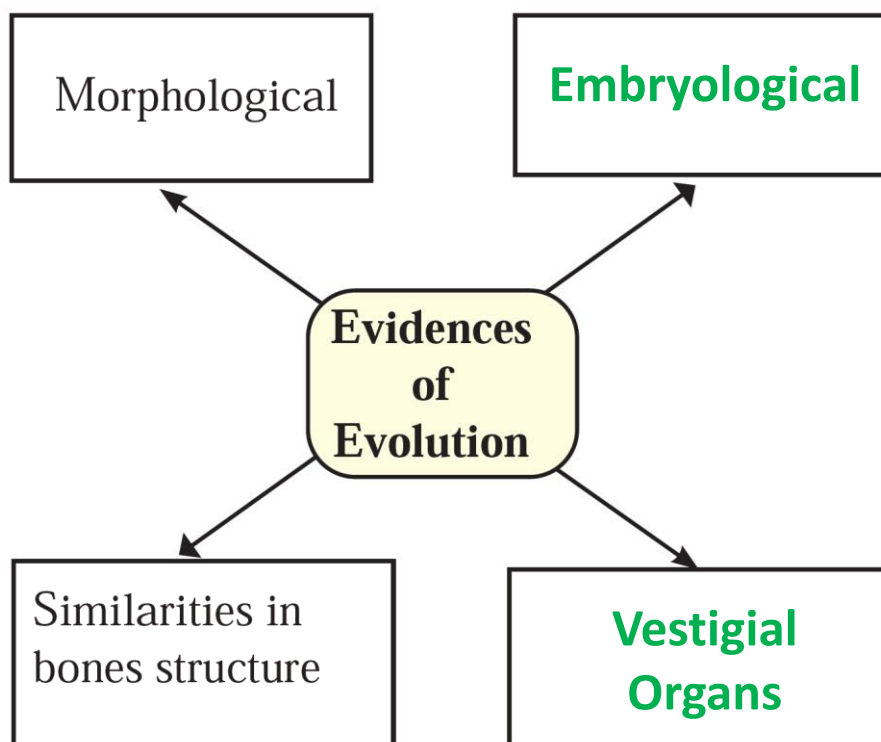
Chapter 1



Heredit & Evolution



1. Complete the following diagram.



2. Read the following statements and justify the same in your own words with the help of suitable examples.

a. Human evolution began approximately 7 crore years ago.

1. Last dinosaurs disappeared approximately 7 crore years ago.
2. Ancestors of monkey-like animals where lemur-like animals which evolved during this time.
3. The tails of these monkey-like creatures started disappearing very gradually around 4 crore years ago.
4. The body and brain both increased in volume which formed the first ape-like animals.
5. Later, the human evolution took place by changes in the brain volume, the ability to walk upright, excessive use of hand for manipulation.
6. This journey of human evolution began 7 crore years ago. But the true wise and intelligent man arose around 50,000 years ago.

b. Geographical and reproductive isolation leads to speciation.

1. Species is the group of organisms that can produce fertile individuals through natural reproduction.
2. Each species grows in specific geographical condition. Their food, habitat, reproductive ability and period is different.
3. Along with genetic variation, geographical and reproductive changes are responsible for formation of new species.
4. Hence, geographical and reproductive isolation leads to speciation.

c. Study of fossils is an important aspect of study of evolution.

1. Large number of organisms get buried due to disasters like flood, earthquake, volcano, etc.
2. Remnants and impressions of such organisms remain preserved underground. These are called as fossils.
3. With the help of carbon-dating technique, it is easy to determine the age of fossils.
4. Once the age of fossil has been determined, it becomes easy to deduce information about other **erstwhile** organisms.
5. Hence, study of fossils is an important aspect of study of evolution.

Erstwhile = earlier

d. There is evidences of **fatal** science among chordates.

1. Very young embryos of fish, amphibian, reptiles, birds and mammals show extreme similarities in the early stages .
2. Those similarities decrease gradually.
3. The initial similarity between the vertebrate embryo's is an evidence that during evolution there was a common ancestor for all the vertebrate classes.
4. This is called embryological evidence for vertebrate evolution.

In textbook it is fatal. It should be **fetal**. (Fetus = unborn offspring)

3. Complete the statements by choosing correct options from bracket.

(Gene, Mutation, Translocation, Transcription, Gradual development, Appendix)

- a. The causality behind the sudden changes was understood due to mutation principle of Hugo de Vries.
- b. The proof for the fact that protein synthesis occurs through gene was given by George Beadle and Edward Tatum.
- c. Transfer of information from molecule of DNA to mRNA is called as Transcription process.
- d. Evolution means Gradual development.
- e. Vestigial organ Appendix present in human body is proof of evolution.

4. Write short notes based upon the information known to you.

a. Lamarckism

1. Jean-Baptiste Lamarck proposed that morphological changes occurring in living organisms are responsible for evolution and the reason behind those morphological changes is activities or laziness of that organism.
2. The neck of giraffe has become too long due to browsing on leaves of tall plants by extending their neck for several generations.
3. Shoulders of ironsmith have become very strong due to frequent hammering movements.
4. Wings of birds like emu and ostrich have become weak due to no use.
5. Legs of the birds like swan and duck have become useful for swimming due to living in water.
6. Snakes have lost their legs by modification in their body for burrowing habit.
7. All these examples are types of acquired characters and are transferred from one generation to another This is called as theory of inheritance of acquired characters or Lamarckism.
8. Development of organs due to specific activities or their degeneration due to no use at all was widely accepted but transfer of those characters from generation to generation was rejected.

9. Because it had been verified many times that modifications brought in us are not transferred to next generation and thereby Lamarck's theory was disproved.

b. Darwin's theory of natural selection.

1. Charles Darwin had collected innumerable specimens of plants and animals and depending upon the observations of those specimens; he published the theory of natural selection which preaches the survival of fittest.
2. For this purpose, Darwin had published a book titled '**Origin of Species**'.
3. While explaining the concept, Darwin says that all the organisms reproduce prolifically.
4. All the organisms compete in a life-threatening manner.
5. In this competition, only those organisms sustain which show the modifications essential for winning the competition.
6. However, besides this, natural selection also plays important role because nature selects only those organisms which are fit to live, and the rest perish.
7. Sustaining and selected organisms can perform reproduction and thereby give rise to the new species with their own specific characters.
8. Darwin's theory of natural selection was widely accepted for long duration. However, some objections were raised against the theory.

9. Some of the main objections are-

- i. Natural selection is not the only factor responsible for evolution.
- ii. Darwin did not mention any explanation about useful and useless modifications.
- iii. There is no explanation about slow changes and abrupt changes.

10. Irrespective of all these objections, Darwin's work on evolution has been a milestone.

c. Embryology

1. Embryology is the science which deals with the study of development of an organism from an embryo.
2. Comparative study of embryonic developmental stages of various vertebrates shows that all embryos have extreme similarities during initial stages.
3. Those similarities decrease gradually.
4. Similarities in initial stages indicate the common origin of all these animals.

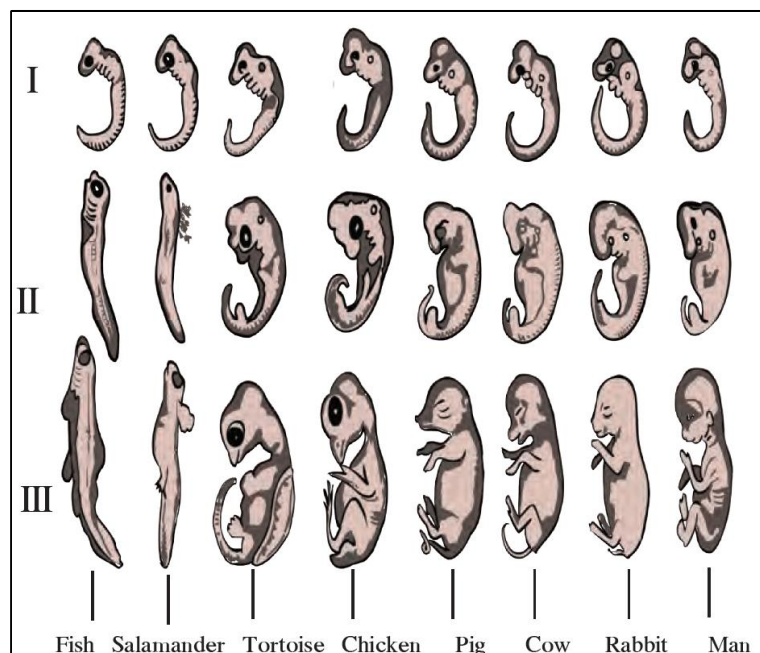


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d. Evolution.

1. Evolution is a gradual change occurring in living organisms over a long duration.
2. This is a very slow-going process through which development of organisms is achieved.
3. Changes in various components ranging from stars and planets in space to the biosphere present on the earth should be included in the study of evolution.
4. Formation of new species due to changes in specific characters of several generations of living organisms as a response to natural selection is called as evolution.
5. Different theories about origin and evolution of life have been proposed till today of which theory of gradual development of living organisms is accepted.
6. According to this theory, first living material (Protoplasm) has been formed in the ocean. In due course of time, unicellular organism was formed.
7. Gradual changes took place in unicellular organisms from which larger and more complex organisms were formed. Duration of all these changes is about 300 crore years.

8. Changes and development in living organisms has been all round and multi-dimensional and this led to evolution of different types of organisms.
9. Hence, this overall process is called as evolution which is organizational.
10. Progressive development of plants and animals from the ancestors having different structural and functional organization is called evolution.

e. Connecting links

1. Some plants and animals show some morphological characters by which they can be related to two different groups; hence, they are called connecting links.
2. In Peripatus, characters like segmented body, thin cuticle and parapodia are present like Annelida. Similarly, these animals show tracheal respiration and open circulatory system like arthropods. This indicates that Peripatus is the connecting link between Annelida and Arthropoda.
3. Similarly, Duck Billed Platypus lays eggs like reptiles but shows relationship with mammals too due to presence of mammary glands and hair.
4. Lung fish performs respiration with lungs irrespective of being a fish.
5. These examples indicate that mammals have evolved from reptiles and amphibians from fishes.

5. Define heredity. Explain the mechanism of hereditary changes.

1. Heredity is the transfer of biological characters from one generation to another via genes.
2. Living organisms can produce new individuals like themselves due to genes only and some of those genes are transmitted to the next generation without any changes.
3. Due to this, some of the characters of parents are transmitted to their offsprings.
4. However, sometimes sudden changes occur in those genes.
5. Sometimes any nucleotide of the gene changes its position and that causes a minor change which is called as mutation.
6. Some mutations may be minor, but some may be considerable.
7. Mutation may cause the genetic disorder like sickle cell anaemia.
8. Also, at the time of meiosis the crossing over takes place this creates new recombination of the genetic information therefore the haploid gametes produced carry changed hereditary characters.

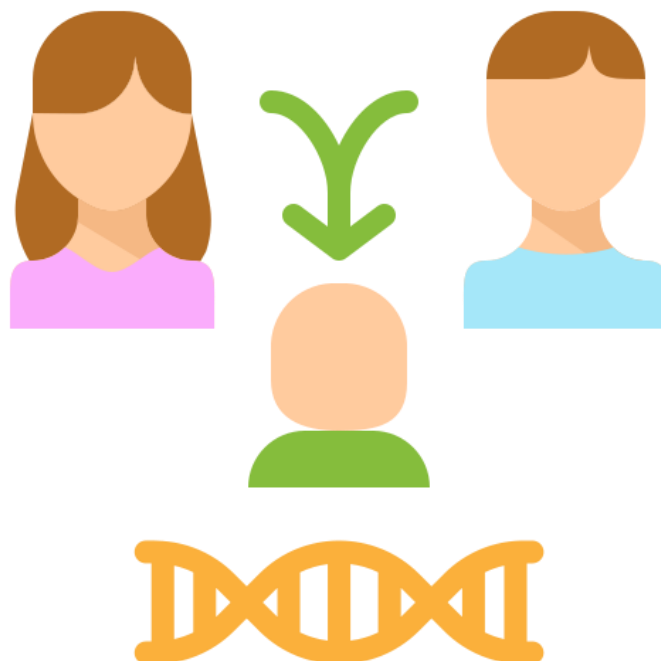
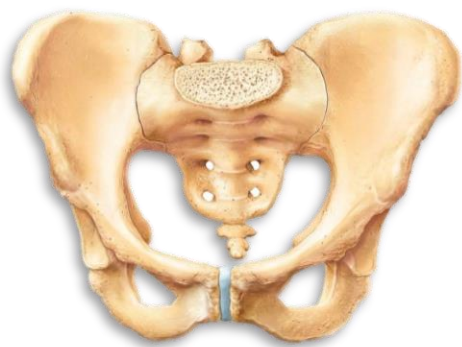


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6. Define vestigial organs. Write names of some vestigial organs in human body and write the names of those animals in whom same organs are functional.

1. Degenerated or underdeveloped useless organs of organisms are called as vestigial organs.
2. Various vestigial organs like tail-bone (coccyx), wisdom teeth, and body hair are present in body of human being.
3. Appendix is a vestigial organ in humans, but is fully functional in ruminants.
4. Muscles of ear pinna are vestigial in humans but are useful in monkey for movement of ear pinna.



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7. Answer the following questions.

a. How are the hereditary changes responsible for evolution?

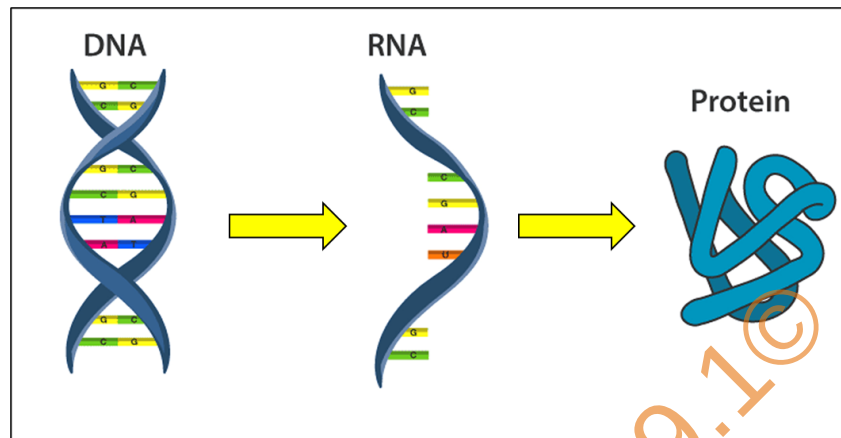
1. Formation of new species due to changes in specific characters of several generations of living organisms as a response to natural selection is called as evolution.
2. At present, crores of species of plants, range from Chlorella to Banyan tree and animals ranging Amoeba to giant whale are existing on earth.
3. First primitive cell may have been formed from the mixture of different types of organic and inorganic compounds.
4. Due to mutation, there may have been some differences among these cells and according to the principle of natural selection, some may have shown good growth and some may have perished which could not adjust with surrounding.
5. Sustaining and selected organisms perform reproduction and thereby, give rise to new species with their own specific characters.
6. Thus, we can say that hereditary changes and natural selection are responsible for evolution.



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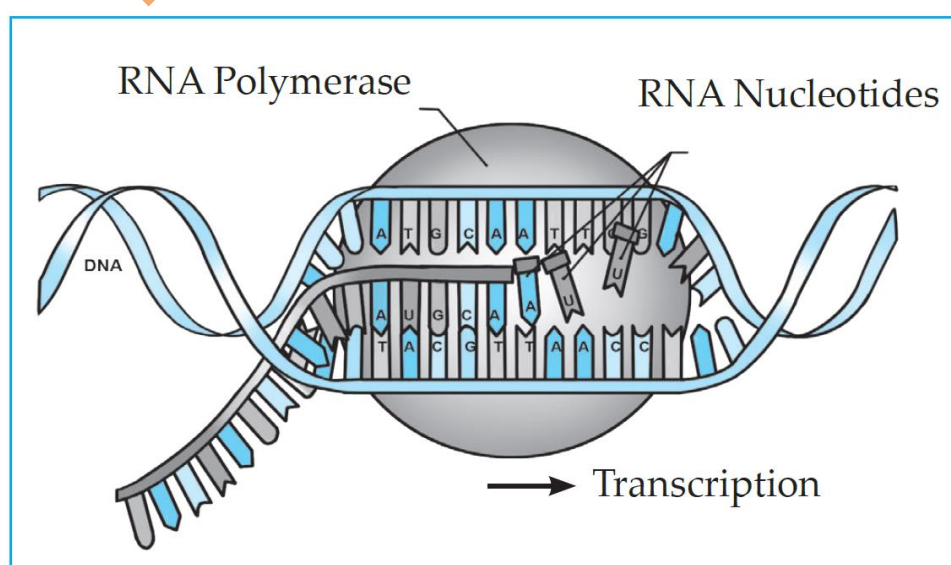
b. Explain the process of formation of complex proteins.

1. Information about protein synthesis is stored in the DNA. Synthesis of appropriate proteins as per requirement is necessary for the body.
2. These proteins are synthesized by DNA through the RNA. This is called as The **Central Dogma** of life.

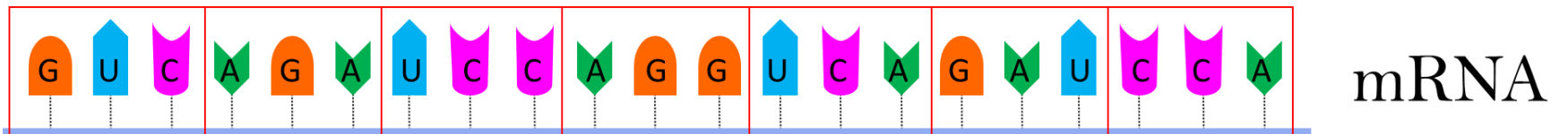


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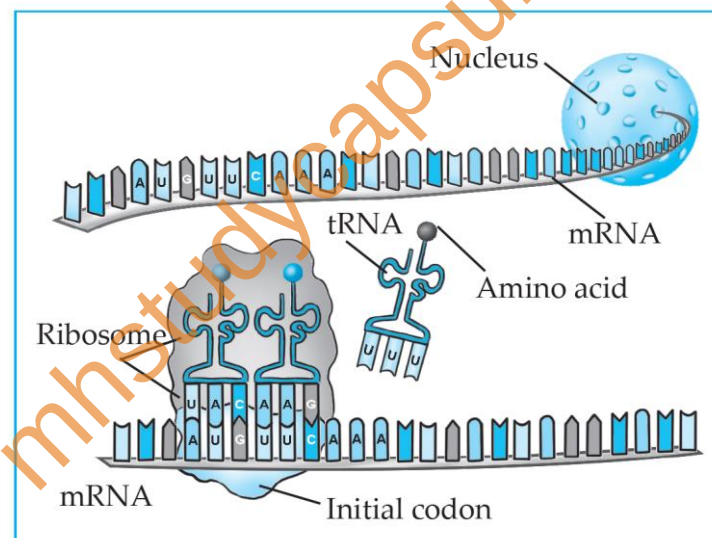
3. mRNA is produced as per the sequence of nucleotides on DNA. Only one of the two strands of DNA are used in this process.
4. The sequence of nucleotides in mRNA being produced is always complementary to the DNA strand used for synthesis. This process of RNA synthesis is called transcription.



5. The mRNA formed in the nucleus comes in cytoplasm. It brings in the coded message from DNA.
6. The message contains the codes for amino acids. The code for each amino acid consist of three nucleotides. It is called as triplet codons.
7. mRNA is made up of thousands of triplet codons.



8. As per the message on mRNA, amino acids are supplied by the tRNA.
9. For this purpose, tRNA has 'anticodon' having complementary sequence to the codon on mRNA. This is called as 'translation'.



10. The amino acids brought in by tRNA are bonded together by peptide bonds with the help of rRNA.
11. During this process, the ribosome keeps on moving from one end of the mRNA to the other end by the distance of one triplet codon. This is called as 'translocation'.
12. Many such chains come together to form complex proteins.

c. Explain the theory of evolution and mention the proof supporting it.

1. According to the theory of evolution, first living material (protoplasm) has been formed in ocean.
2. In due course of time, unicellular organism was formed.
3. Gradually, changes occurred in the unicellular organism from which larger and more complex organisms were formed.
4. All those changes were slow and gradual. Duration of all these changes is about 300 crore years.
5. Changes and development in living organisms has been all round and multi-dimensional and this led to evolution of different types of organisms.
6. Hence, this overall process is called as evolution which is organizational.
7. Progressive development of plants and animals from the ancestors having different structural and functional organization is called evolution.
8. To support the theory of evolution, various types of evidences are available.
9. Morphological evidences support the theory. Various similarities like structure of mouth, position of eyes, structure of nostrils and ear pinnae and thickly distributed hairs on body are seen in animals whereas similarities in characters like leaf shape, leaf venation, leaf petiole, etc. occur in case of plants.

This indicates that there are some similarities in those groups and hence it proves that their origin must be same and must have common ancestors.

10. Anatomical evidences like the bone structure of a human arm, a bull's leg, a bat's wing and a whale fin have similar structures attributing that they have developed from common ancestors.
11. Vestigial organs like tail bone or coccyx present in humans indicate the link between humans who have lost a tail and monkeys who have a tail.
12. At embryological level, similarities are observed during initial stage which indicates common origin.
13. Thus, there are a few evidences which support the fact that evolution is sequential.

d. Explain with suitable examples importance of anatomical evidences in evolution.

1. There doesn't seem any superficial similarity between a human hand, a cat's foreleg, flipper of a whale and patagium of a bat.
2. Use of each of these structures is different in respective animals.
3. However, there is a similarity in structure of bones and bony joints in organs of each of these animals.
4. This similarity indicates that those animals may have common ancestor.

e. Define fossil. Explain importance of fossils as proof of evolution.

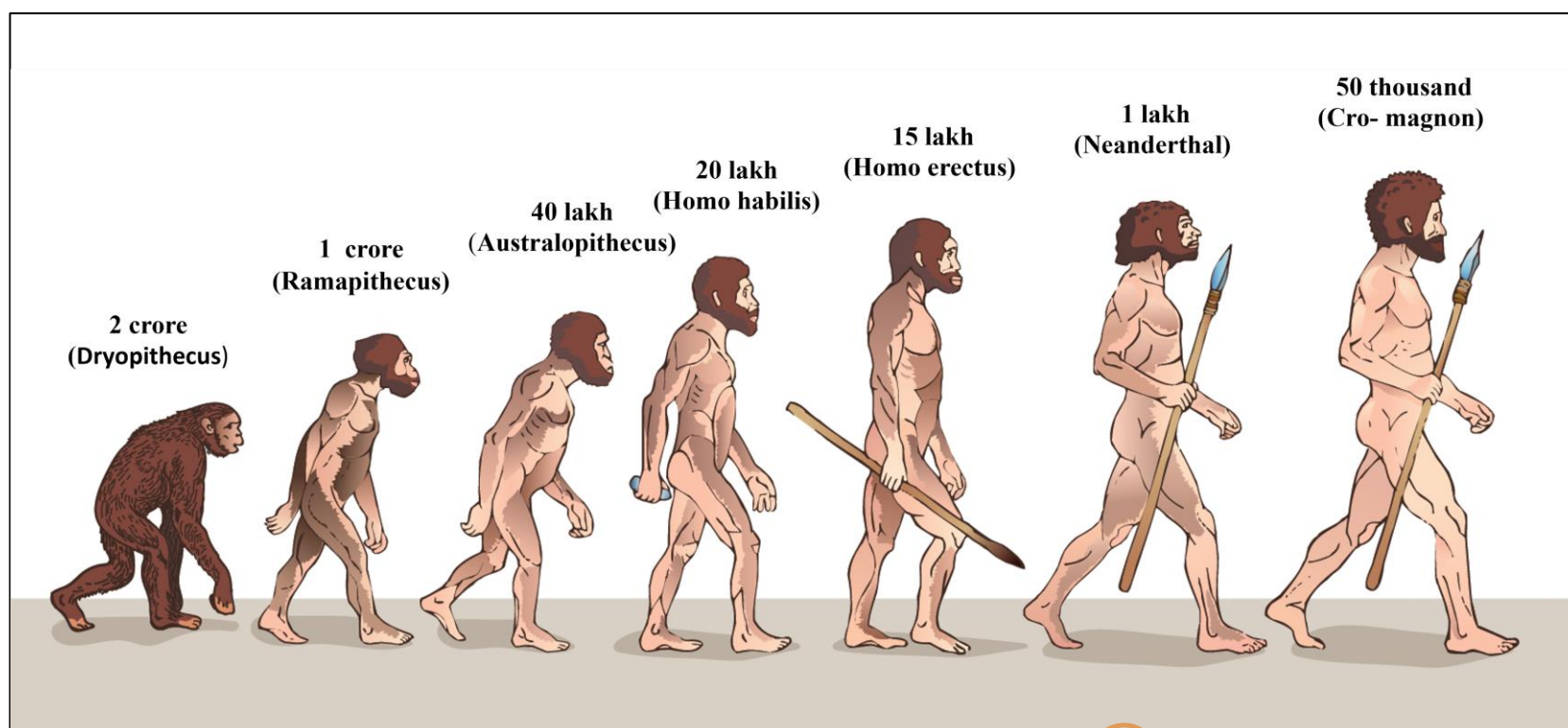
1. Remnants and impressions of organisms which get buried remain preserved underground. These are called as fossils. Study of fossils is an important aspect of study of evolution.
2. Carbon consumption of animals and plants stops after death and since then, only the decaying process of C-14 occurs continuously.
3. In case of dead bodies of plants and animals, instead of remaining constant, the ratio between C-14 and C-12 changes continuously as C-12 is non-radioactive.
4. The time passed since the death of a plant or animal can be calculated by measuring the radioactivity of C-14 and the ratio of C-14 to C-12 present in their body.
5. This is carbon dating method. It is used in determining the age of human fossils and manuscripts.
6. Once the age of fossil has been determined, it becomes easy to deduce information about other erstwhile organisms.
7. It seem that vertebrates have been slowly originated from invertebrates.



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f. Write evolutionary history of modern man.

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1. Approximately seven crore years ago, monkey-like animals are said to be evolved from some ancestors who were more or less like the modern lemurs.
2. Tail of these monkey-like animals of Africa is said to have disappeared about 4 crore years ago. They developed due to enlargement in brain their hands were also improved, and thus ape-like animals were evolved.
3. Evolution of some of the 2 crore years old species of apes seems to have occurred in different ways. They had to use their hands more for eating food and other work. Those apes started to live on land as the forests started to decline due to dry environment. Their pelvic girdle developed in such a way that they started to stand in erect posture in grasslands and thereby their hands became available for use anytime.

4. First human like animal recorded was 'Ramapithecus' ape from East Africa.
5. This ape grew up in size and became more intelligent and thus ape of South Africa evolved about 40 lakh years ago.
6. The morphology of these human-like animals started to appear like the members of the genus Homo, about 20 lakh years ago and thus skilled human developed.
7. About 15 lakh years ago, humans walking with erect posture evolved and existed in China and Indonesia of Asian continent.
8. Evolution of upright man continued in the direction of developing its brain for the period of about 1 lakh years and meanwhile it discovered fire.
9. Brain of 50-thousand-year-old man had been sufficiently evolved to the extent that it could be considered as member of the class-wise-man (Homo sapiens).
10. Neanderthal man can be considered as the example of wise-man.
11. The Cro-Magnon man evolved about 50 thousand years ago and afterwards, this evolution had been faster than the earlier.

INTEXT QUESTIONS

1. Which component of the cellular nucleus of living organisms carries hereditary characters?

Cellular nucleus of living organisms contains the genetic material like DNA, which carries hereditary characters.

2. What do we call to the process of transfer of physical and mental characters from parents to the progeny?

The process of transfer of physical and mental characters from parents to the progeny is called heredity.

3. Which are the components of DNA molecule?

DNA has three types of chemical components:

- (1) phosphate
- (2) a sugar called deoxyribose
- (3) four nitrogenous bases-adenine, guanine, cytosine and thymine.



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4. Explain the meaning of genetic disorders and give names of some disorders.

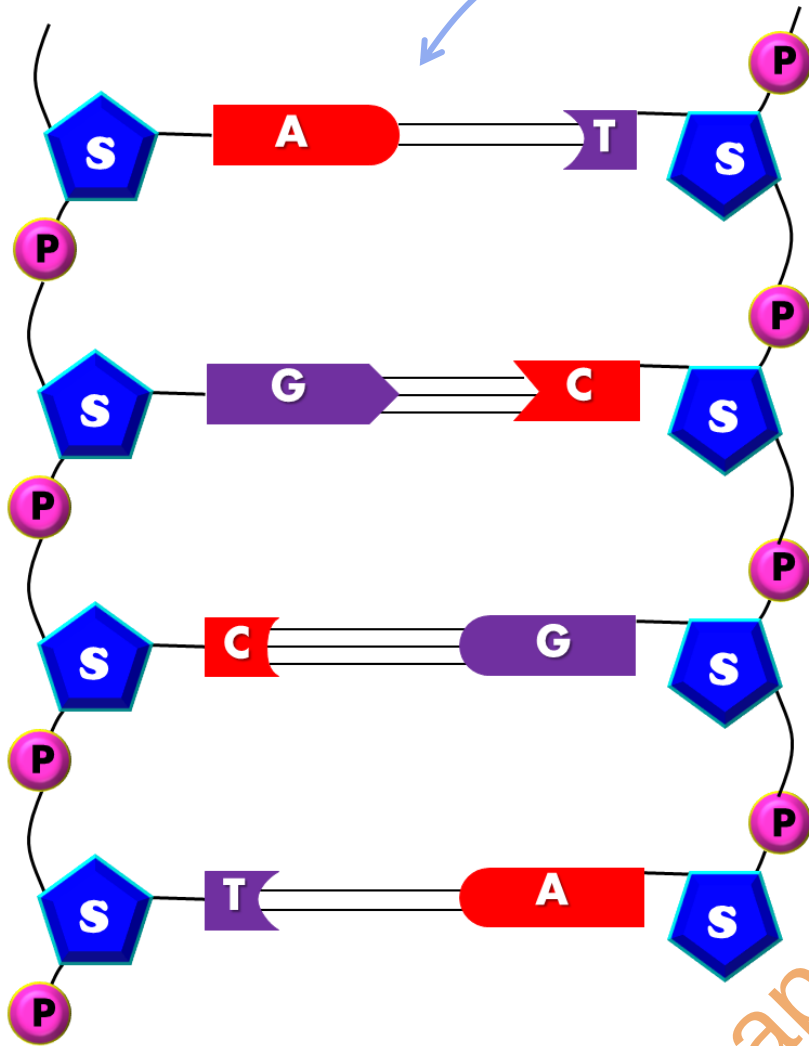
1. Genetic disorders are disorders caused due to mutations in the gene. It is present from birth.
2. Diabetes, obesity, infertility, cleft palate, heart disease, hemophilia, sickle cell anaemia are some of the genetic disorders.

5. Sketch and explain the structure of DNA and various types of RNA.

1. Each strand in the molecule of DNA is made up of many small molecules known as 'nucleotide'.
2. In the structure of a nucleotide, a molecule of a nitrogenous base and phosphoric acid are each joined to a molecule of sugar.
3. There are four types of nitrogenous bases adenine, guanine, cytosine and thymine.
4. Adenine and guanine are called as 'purines' while cytosine and thymine are called 'pyrimidines'.
5. As there are four types of nitrogenous bases, nucleotides also are of four types.
6. Adenine always pairs with thymine by two hydrogen bonds.
7. Cytosine always pairs with guanine by three hydrogen bonds.

8. The two strands of DNA are coiled around each other, forming a double helix structure.

Diagram to be drawn in exam



➤ Various types of RNA.

According to function, there are three types of RNA.

1. Ribosomal RNA (rRNA) : The molecule of RNA which is a component of the ribosome organelle is called a ribosomal RNA. Ribosomes perform the function of protein synthesis.
2. Messenger RNA (mRNA) : The RNA molecule that carries the information of protein synthesis from genes in the cell nucleus to ribosomes in the cytoplasm which produce the proteins, is called messenger RNA.

3. Transfer RNA (tRNA) : The RNA molecule which, according to the message of the mRNA carries the amino acid up to the ribosomes is called transfer RNA.

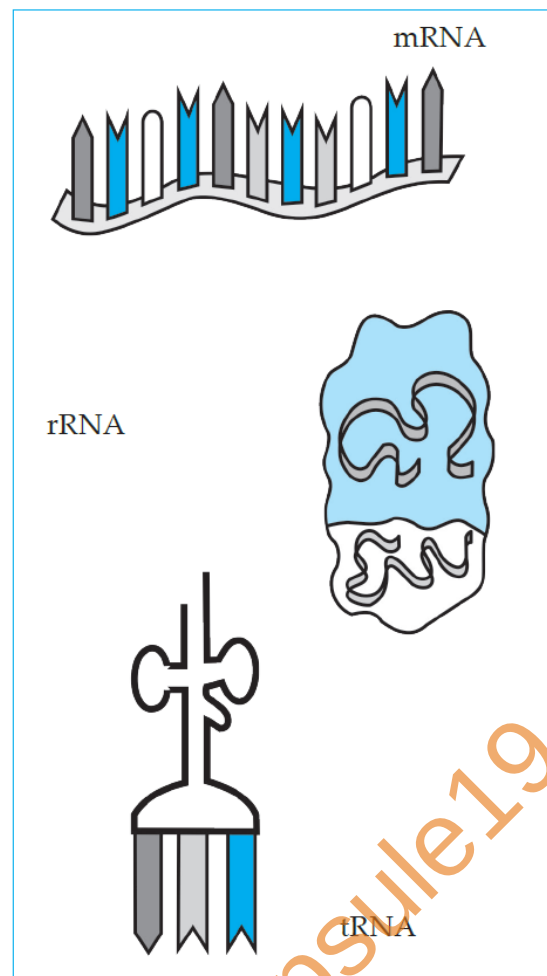
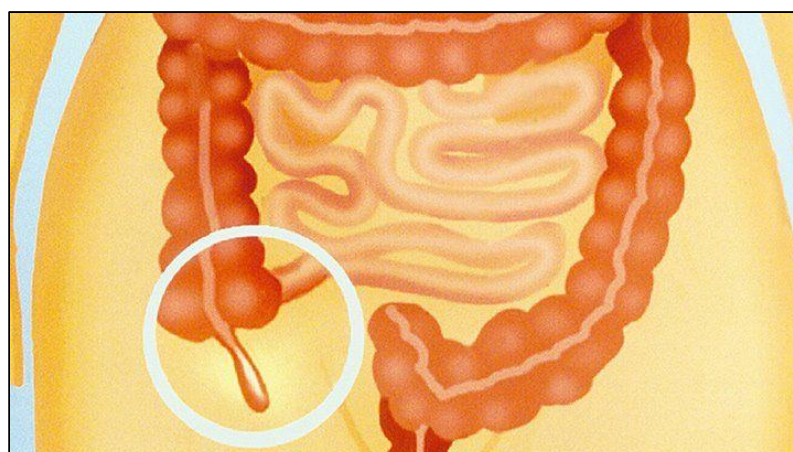


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6. What is the function of the appendix of our digestive system?

1. The appendix is a vestigial organ that was once used by our herbivorous ancestors.
2. It was found that in herbivorous vertebrates the appendix is larger, and it helped in the digestion of tough fibrous food such as the bark of a tree.
3. It is also supposed to be a store house of good bacteria.



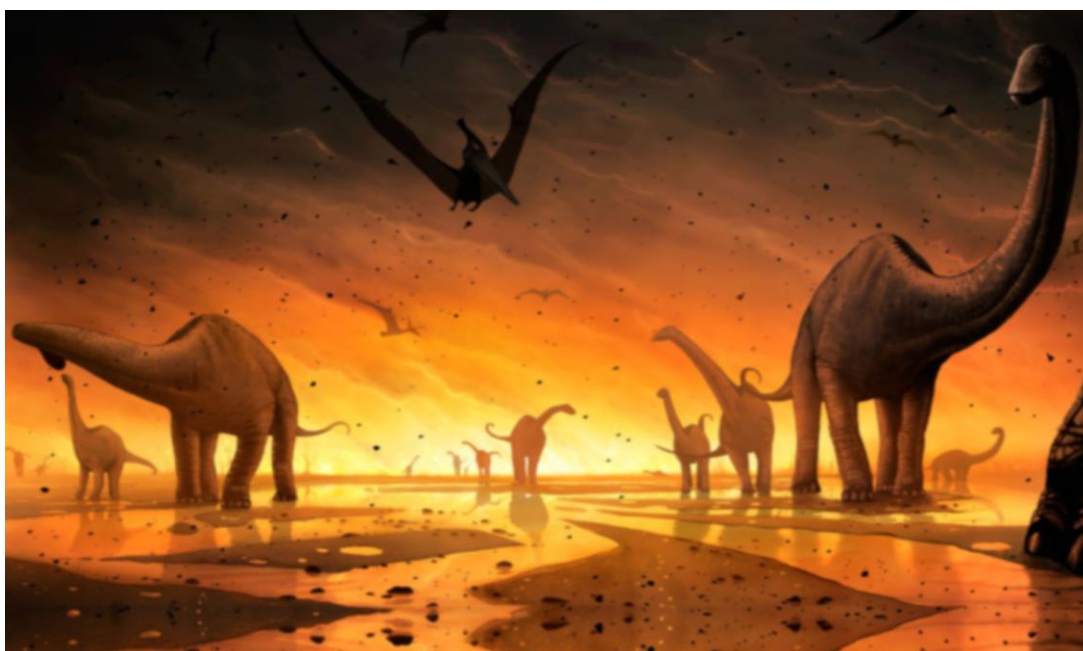
7. Are our wisdom teeth really useful for chewing the food?

No, people do not need wisdom teeth now for chewing the food but earlier they were used for chewing tough plants which were important part of the human diet.

8. Why did the huge animals like dinosaur become extinct?

Inspite of the fact that the cause for dinosaur extinction is still a mystery, following events could have played a role.

1. Gas seeping and ash evolving from volcanoes suffocated most of the dinosaurs.
2. The entire population of dinosaur may have been wiped out due to diseases.
3. The dinosaurs could not survive after a big meteorite crashed into earth, drastically changing the climatic conditions.
4. Imbalances in food chain led to starvation of the dinosaurs.



9. Why are many species of animals and birds getting extinct?

1. Animals and birds' extinction may be caused by natural occurrences such as climatic heating or cooling or changes in sea levels.
2. Habitat destruction as farming land expands and forests are cut-down. This is the main cause along with pollution and overfishing or hunting.

10. Which are the different organs in body of organisms?

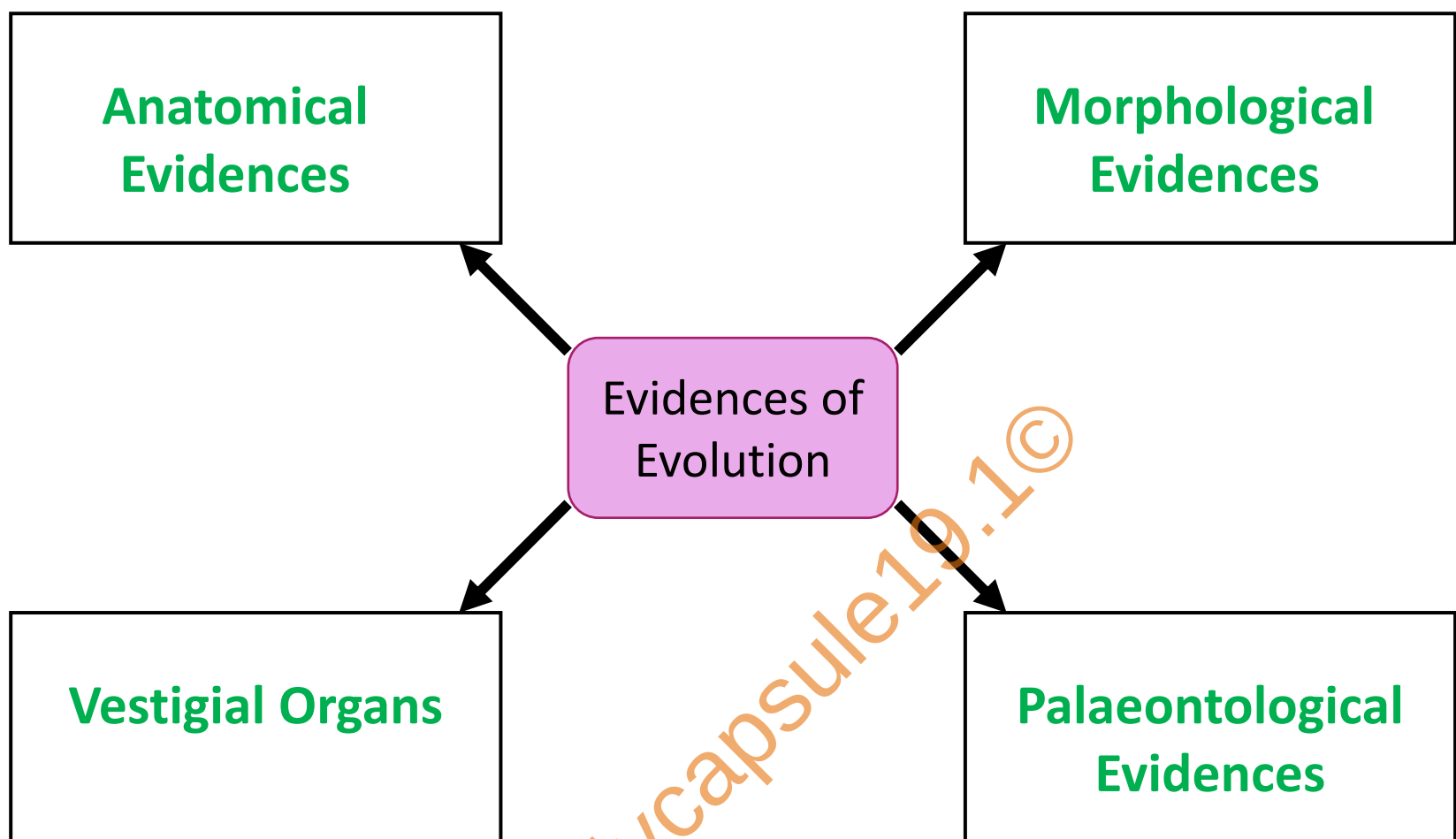
1. In digestive system - (Mouth, Stomach, Liver, Pancreas, intestines, etc.)
2. Skeletal system - (Bones)
3. Circulatory system - (Heart, Arteries, Veins, etc.)
4. Respiratory system - (Diaphragm, Lungs, Larynx, etc) are some of the organs in an organism.

11. Is each of the organs useful to organism?

1. Most of the organs are useful to organisms. However, few organs that are present now have become vestigial i.e., lost their function in the process of evolution.
2. In human beings, appendix, tail-bone (coccyx), wisdom teeth and body hair are not useful now.

Previous Year Questions

1. Complete the following chart. (2M)



2. Answer the following Questions: (3M)

1. What do you mean by central dogma?
2. What is transcription?
3. What is meant by triplet codon?

1. Information about protein synthesis is stored in the DNA and synthesis of appropriate proteins as per requirement is necessary for body. These proteins are synthesized by DNA through the RNA. This is called as 'Central Dogma'.

2. mRNA is produced as per the sequence of nucleotides on DNA. Only one of the two strands of DNA is used in this process. The sequence of nucleotides in mRNA being produced is always complementary to the DNA strand used for synthesis. Besides, there is uracil in RNA instead of thymine of DNA. This process of RNA synthesis is called as 'transcription'.
3. The code for each amino acid consists of three nucleotides. It is called as 'triplet codon'.

3. Define vestigial organs. Write any two names of vestigial organs in human body. (2M)

Same answer as question 6 of exercise.

4. Explain with suitable examples importance of anatomical evidences in evolution. (3M)

Same answer as question 7-d of exercise.

5. Explain Darwin's theory of natural selection. (3M)

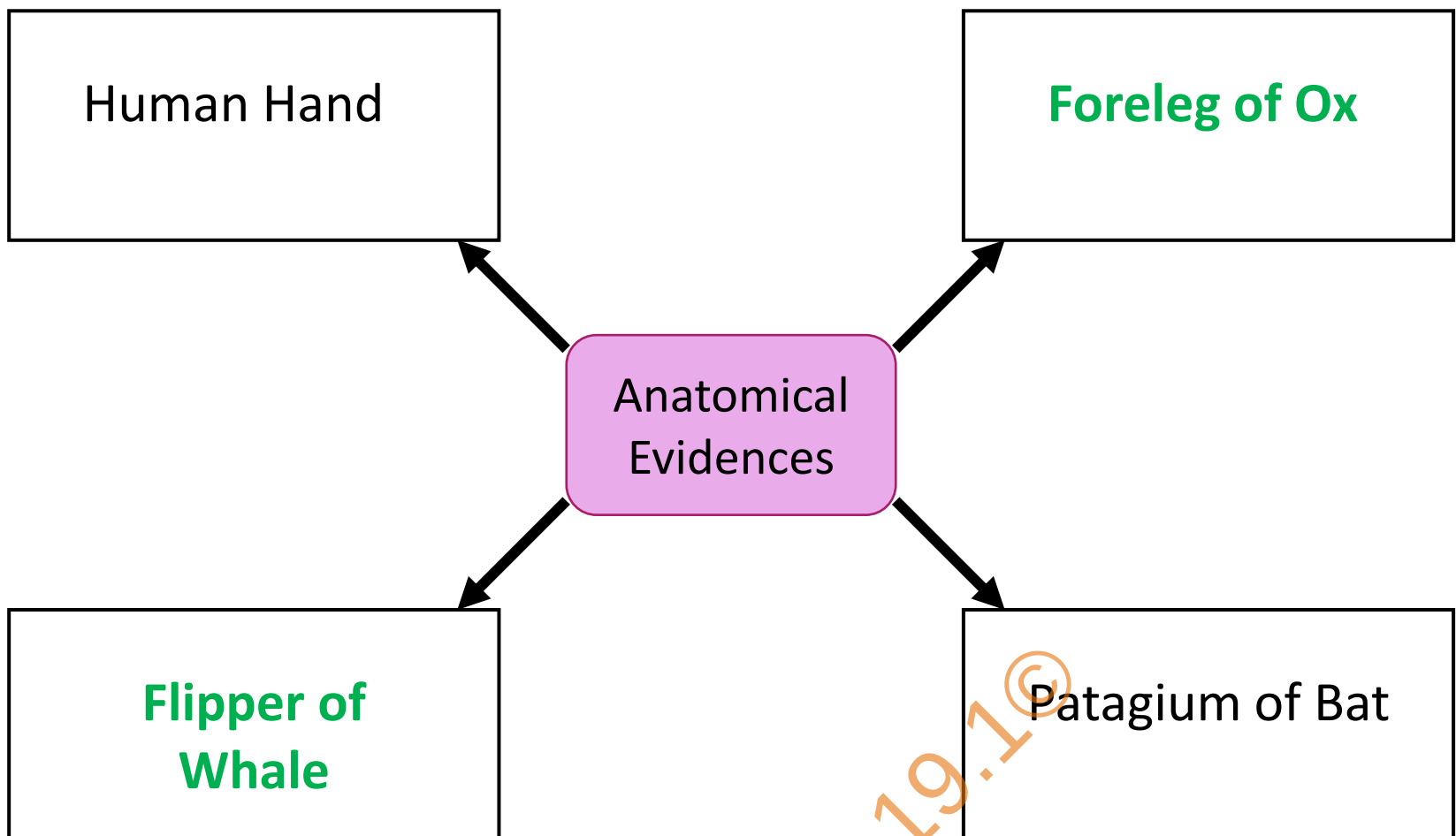
Same answer as question 4-b of exercise.

6. a. Define Vestigial Organs. (3M)

b. Write name of any two vestigial organs in human body.

c. Explain how one human vestigial organ is functional in another animal. Same answer as question 6 of exercise

7. Complete the following diagram. (2M)



8. What do you mean by fossils? Write the importance of fossils.

Write the name of method to find the age of fossils. (3M)

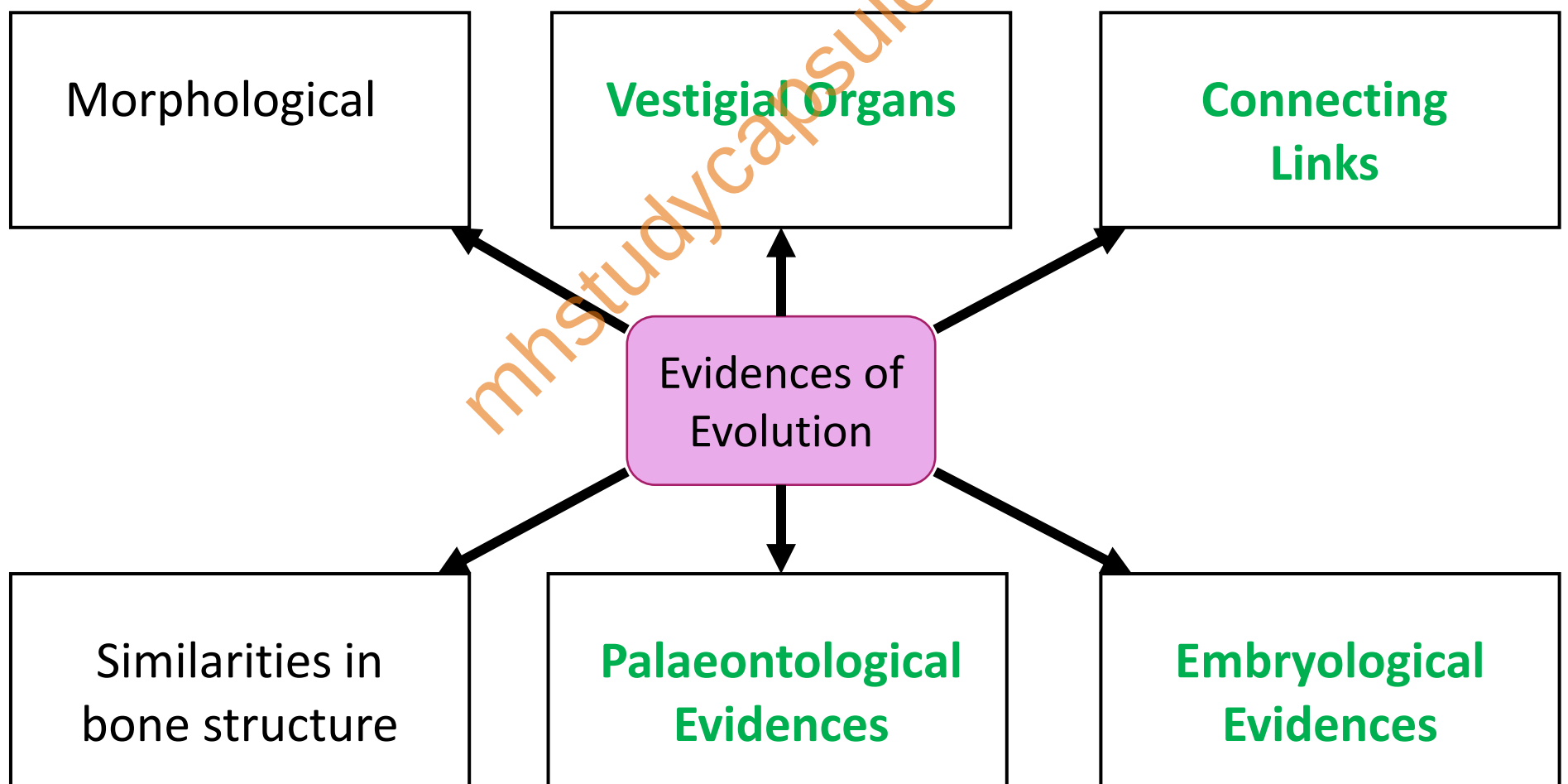
Same answer as question 7-e of exercise.

9. Write the objections raised against Darwin's theory of natural selection. (3M)

1. Charles Darwin had collected innumerable specimens of plants and animals and depending upon the observations of those specimens; he published the theory of natural selection which preaches the survival of fittest.
2. For this purpose, Darwin had published a book titled 'Origin of Species'.

3. Darwin's theory of natural selection was widely accepted for long duration.
4. However, some objections were raised against the theory. Some of the main objections are-
 - i. Natural selection is not the only factor responsible for evolution.
 - ii. Darwin did not mention any explanation about useful and useless modifications.
 - iii. There is no explanation about slow changes and abrupt changes.
5. Irrespective of all these objections, Darwin's work on evolution has been a milestone.

10. Complete the following diagram. (2M)



Vestigial organ _____ present in human body is proof of evolution.

(A) Human hand

(B) Appendix

(C) Ear

(D) Tail

Wisdom Teeth

Transfer of information from molecule of DNA to mRNA is called _____ process.

(A) Transcription

(B) Translation

(C) Translocation

(D) Mutation

I am connecting link between Reptilia and mammals. What is my name?

Duck billed platypus